

16. Compare different types Classification Algorithms and evaluate their performance.

Aim

To compare different **classification algorithms** and evaluate their performance using accuracy.

Algorithms Used

- Logistic Regression
- K-Nearest Neighbours
- Decision Tree
- Naive Bayes
- Support Vector Machine

Software Required

- Python 3.x
- Scikit-learn
- Pandas

Procedure

1. Load the Iris dataset.
2. Divide the dataset into training and testing sets.
3. Create different classification models.
4. Train each model.
5. Predict the test data.
6. Calculate accuracy for each model.
7. Compare the results.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
```

```

from sklearn.svm import SVC

from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

models = {
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Naive Bayes": GaussianNB(),
    "SVM": SVC()
}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print(name, ":", accuracy)

```

Output:

```

Logistic Regression : 1.0
KNN : 1.0
Decision Tree : 1.0
Naive Bayes : 1.0
SVM : 1.0

```

Result

Thus, different classification algorithms were successfully implemented and compared based on their accuracy.